

Telescreen Project Concept (Positive Family Communication Platform)

Purpose

Telescreen is a family communication system designed to help relatives stay connected with elderly family members, particularly those living alone or suffering from conditions such as Parkinson's disease. The name intentionally reuses the historical term 'Telescreen' but for the opposite purpose: to improve quality of life, independence and family communication rather than surveillance.

Problem Statement

Many elderly users find modern television interfaces and remote controls difficult to operate. Starting a video call often requires multiple remote-control actions: turning on the television, navigating menus, launching the video application and answering the incoming call. For users with Parkinson's disease or reduced dexterity, these steps can become impossible without assistance.

Proposed Solution

A Samsung Z2 (Tizen 2.4) smartphone acts as a dedicated home automation controller. The Z2 continuously monitors incoming SMS messages from authorised family members. When a recognised command is received, it sends Android Debug Bridge (ADB) commands over the local network to the television.

Typical Workflow

1. Family member sends an authorised SMS command.
2. Samsung Z2 receives the SMS.
3. Z2 wakes the television (using the available wake mechanism such as network wake or other supported method).
4. Z2 connects to the television over ADB.
5. Running applications are closed if necessary.
6. The TrueConf application is launched automatically.
7. Incoming TrueConf calls are configured to auto-answer.
8. The family member can immediately speak with the senior citizen without requiring any interaction with the television remote.

System Components

- Samsung Z2 (Tizen 2.4) running the Telescreen controller application.
- Android-based television with wireless ADB enabled and available while in network standby.
- Television connected to mains power.
- Television retaining network connectivity while in standby.
- Logitech C710 webcam providing camera, microphone and speakers (or equivalent compatible USB webcam).

Technical Assumptions

The proposed solution depends on:

- The television supporting wireless ADB.
- Network connectivity remaining active while the television is in standby.
- The television remaining connected to power.
- The television allowing applications such as TrueConf to be launched via ADB shell commands.

Future Enhancements

- HDMI-CEC wake-up support.
- Wake-on-LAN where supported.
- Vendor-specific TV APIs.
- Encrypted SMS commands.
- Remote status reporting.
- Portable MiniADB library shared across Tizen, Linux, Windows and Raspberry Pi.